Influence of Simulated Weightlessness on the Oral Pharmacokinetics of Acetaminophen as a Gastric Emptying Probe in Man: A Plasma and a Saliva Study

Peggy Gandia, PharmD, Marie-Pierre Bareille, PharmD, Sylvie Saivin, PharmD, PhD, Anne Pavy Le-Traon, MD, PhD, Michel Lavit, PharmD, Antonio Guell, PhD, and Georges Houin, PharmD, PhD

This study evaluated the effect of simulated weightlessness on gastric emptying, using acetaminophen as a probe and –6° head-down bed rest to simulate zero gravity. Eighteen volunteers were given 1 g of acetaminophen orally before the bed rest and at days 1, 18, and 80. C_{max}, t_{max}, $AUC_{0-\infty}$, AUC_{0-t}, and $t_{1/2}$ were calculated for plasma and saliva. The plasma C_{max} showed a significant increase (10.43 μg/mL [day 1] to 14.74 μg/mL [day 80]), while t_{max} significantly decreased (1.41 h [day 1] to 0.91 h [day 80]). Similar results were obtained with saliva, and there were significant increases in the AUCs. The good correlation between the plasma and saliva data suggests that saliva sampling can be valid for acetaminophen pharmacokinetics. The changes in C_{max} and t_{max} indicated more rapid drug absorption, which could have been as a result of faster gastric emptying or an increased blood flow to the intestine.

Keywords: *Acetaminophen pharmacokinetics; bed rest; gastric emptying; simulated weightlessness*
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Since the beginning of the 1960s, humans have embarked on the conquest of space: a new world, but hostile for a number of different reasons. The primary hostile factor is the lack of gravity, principally acting on bones, muscles, and the cardiovascular system. It is likely that modifications of pharmacokinetic disposition occur[1,2] with pharmacological or toxicological consequences.[3] Given the small number of space flights and the difficulties of conducting experiments during missions, earthbound models for simulating weightlessness are used to study the physiological, pharmacokinetic, and pharmacological modifications observed during flights. The most-used model is the –6° head-down bed rest[4] that reproduces bone and muscle losses and the fluid shift from the bottom to the top of the body, observed in space crewmembers.[5,6] In a previous study, we showed changes in lidocaine disposition during a 4-day head-down tilt (–6°). At the end of the experiment, the total clearance of lidocaine increased significantly by 30% when compared with the ambulatory period.[7] Similarly, Kates et al[8] observed an increase in the total clearance of 2% (result not significant) after a 7-day horizontal bed rest when compared with the ambulatory period. In a recent study, we showed that promethazine availability after oral absorption was significantly increased by 30% in simulated weightlessness (–6° head-down tilt) compared with the sitting position (unpublished data). This statistically significant difference can be explained by the effects of posture on drug absorption[9-11]: standing, sitting, or lying on the right side induces a more rapid gastric emptying rate than lying on the left side. Gastric emptying is one of the main factors responsible for the

From Laboratoire de Pharmacocinétique et Toxicologie Clinique, Hôpital Rangueil-Larrey, Toulouse, France (Dr. Gandia, Dr. Saivin, Dr. Lavit, Dr. Houin) and Medes, Clinique Spatiale, Hôpital Rangueil-Larrey, Toulouse, France (Dr. Bareille, Dr. Pavy le-Traon, Dr. Guell). Financial support: Centre National d'Etudes Spatiales, contract numbers 793/2001/CNES/8972 and 2002/CNES/4800000037. Submitted for publication January 27, 2003; revised version accepted July 3, 2003. Address for reprints: Georges Houin, PharmD, PhD, Laboratoire de Pharmacocinétique et Toxicologie Clinique, Hôpital Rangueil-Larrey, TSA 50032, 1 avenue Jean Poulhès, 31059 Toulouse cedex 9, France.
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variability of drug absorption with variations of the C_{max}, t_{max}, and AUC.[1] When a drug is orally administered, the time necessary to pass through the pylorus depends on the presence or absence of food (solid and liquid content) in the stomach and the composition of the food (protein, carbohydrate, lipid).[12-14] After administration, the drug is dispersed in solid or liquid gastric content depending on its chemical and physical properties. Given the diversity of the meals ingested every day by humans, it is impossible to determine exactly the time necessary for a drug to pass through the pylorus, and this phenomenon could be considered as a random event.[15] Modification of peripheral resistance to blood flow is another physiological parameter that could induce change in drug disposition. Variation of the total peripheral resistance induces variation of the total blood flow and, consequently, of the intestinal blood flow, which controls the passive diffusion of many drugs through the intestinal wall.[16] Space conditions are likely to alter the gastric emptying rate due to the effects on gastric reflexes[2] and are well known to modify the peripheral resistance.[17,18] To study these possibilities, we selected the –6° head-down bed rest as a model and acetaminophen as a gastric emptying probe. Gastric emptying was studied using the pharmacokinetics of acetaminophen, which is totally and rapidly absorbed through the intestinal wall by passive diffusion.[19] Consequently, its absorption profile depends directly on gastric emptying.[20,21]

SUBJECTS AND METHODS

Subjects

Eighteen healthy nonsmoking male volunteers were selected for the study. Subjects were excluded if they had acute, chronic, or mental disease; a family history of gastrointestinal disorders; allergy to acetaminophen; drug or alcohol abuse; or abnormalities in standard laboratory tests. Local ethical committee approval was obtained prior the start of the study in accordance with legal requirements, and all subjects gave written consent. The details of the subjects are shown in Table I.

Study Protocol

The study was conducted in the Medes-IMPS (Space Clinic, Rangueil Hospital, Toulouse, France). Gastric emptying was assessed before –6° head-down bed rest (HDT) for baseline data collection in a sitting position for the 6 first hours (HDT0), after which time volunteers could stand up and walk. During head-down bed rest, gastric emptying was studied three times: after 1 day (HDT1), 18 days (HDT18), and 80 days (HDT80). After an overnight fast, the subjects were given 1 g of acetaminophen washed down with 200 mL of water. Since the absorption of acetaminophen directly depends on gastric emptying[21-26] and since the rate of gastric emptying is altered by posture,[9-11,27] during HDT period measurements, the volunteers stayed in the head-down position, lying on their back, for the 6 first hours following administration, after which time they were permitted to change position but always in the supine position. The volunteers were under video camera observation. After 4 hours from acetaminophen administration, the subjects were permitted to imbibe liquids, and after 6 hours, a standard meal was consumed.

Blood was sampled before administration (10 min prior to dosing); at 15, 30, 40, and 50 minutes; and at 1, 1.25, 1.5, 2, 4, 6, 8, 10, and 12 hours after dosing. For the 4 first volunteers of the period HDT0–session 2001, the last sampling time (10 h) fixed in the study protocol was respected. For the other volunteers of the 2001 and 2002 sessions, the last sampling time had to be changed (12 h instead of 10 h) due to timing interactions with other study protocols. Blood samples were collected through a canule inserted into a forearm vein and transferred into lithium heparinized tubes (Vacutainer®, Plymouth, UK).

Saliva was sampled before administration (10 min prior to dosing), at 15 and 30 minutes, and at 1, 1.5, 2, 4, 6, 8, 10, and 12 hours after dosing, without stimulation. As for plasma, the last sampling time was T = 10 hours for the 4 first volunteers of the period HDT0–session 2001 and T = 12 hours for the others. Saliva samples were collected using dental cotton wool cylinders (Starstedt®, Nümbretch, Germany) within 5 minutes before and after each plasma sampling time. Blood and saliva samples were centrifuged (15 min at 3000 rpm), and then the supernatant was frozen at –80°C until assay.

Analytical Methods

Plasma acetaminophen assays were performed using a validated fluorescence polarization immunoassay (FPIA). An analytical validation was performed before starting and during the assays according to the Arlington criteria.[28] A calibration curve ranging from 2 (lower limit of quantification [LLOQ]) to 200 $\mu g \bullet mL^{-1}$ (upper limit of quantification [ULOQ]) was constructed on the first day of the plasma assays, and three levels of quality controls (15, 35, and 150 $\mu g \bullet mL^{-1}$)

Table I Anthropometric Data for the 2001-2002 Volunteers

	Age (Years)	Height (m)	Weight (kg)	Body Mass Index ($kg \times m^{-2}$)
2001 (n = 10)				
Mean	34	1.74	72	23.87
Standard deviation	3	0.05	7	1.63
2002 (n = 8)				
Mean	31	1.75	69	22.47
Standard deviation	3	0.02	5	1.64
Global (n = 18)				
Mean	32	1.75	71	23.25
Standard deviation	4	0.04	7	1.74

were assayed in duplicate for each pharmacokinetic profile (72 profiles).

The determination of acetaminophen concentrations in saliva was performed by a liquid chromatographic method: 50 μL of β-hydroxytheophylline as an internal standard (Sigma, ST Quentin-Fallavier, France) was added to 500 μL of plasma, 1 mL of extraction buffer (1 M potassium dihydrogen-phosphate, 2 M dipotassium hydrogen-phosphate, pH = 7.4), and 5 mL of ethyl acetate. The mixture was shaken vigorously for 10 minutes and then centrifuged for 10 minutes at 2500 rpm at 0°C. Then, 3.5 mL of the supernatant was collected in hemolysis tubes and evaporated under a nitrogen stream at 50°C. The dried residue was dissolved in 300 μL of the mobile phase. The mobile phase was prepared by mixing phosphate buffer, pH = 3.0 (0.005 M potassium dihydrogen-phosphate), and acetonitrile (85/15 (v/v)). The mobile phase was filtered through a nylon membrane filter of 47 mm to 0.2 μm (Nylaflo®, Pall, USA). Then, 50 μL of the dissolved residue was injected into the high-performance liquid chromatography (HPLC) system, which consisted of a 250 × 4mm column Lichrospher 60 RP Select B-C8 (Merck, Darmstadt, Germany) with two columns, as well as a 5-μm 4 × 4mm Lichrospher 60 RP Select B-C8 (Merck, Darmstadt, Germany) at the top and the bottom of the main column. The flow rate was 1 $mL \bullet min^{-1}$, the pressure was 13 bars, and the eluate was monitored by UV detection (Shimadzu, SPD-6A) at 242 nm. The recording apparatus was an integrator PIC 3 (ICS, Toulouse, France). An analytical validation was performed before starting and during the assays according to the Arlington criteria.[28] A calibration curve ranging from 0.1 (LLOQ) to 50 $\mu g \bullet mL^{-1}$ (ULOQ) was constructed, and three levels of quality controls (0.2, 2.5, and 40 $\mu g \bullet mL^{-1}$) were assayed in duplicate every day.

Pharmacokinetic and Statistical Analysis

The interpretation of the drug time-concentration curves for each volunteer and for each HDT (HDT0, HDT1, HDT18, and HDT80) was performed with Kinetica software (InnaPhase®, Champ-sur-Marne, France), using an independent modeling calculation. We measured the following pharmacokinetic parameters: AUC_{0-t}, the area under the plasma concentration versus time curve from 0 to the last measurable concentration (C_{last}), was calculated using the trapezoidal method. $AUC_{0-\infty}$, the area under the plasma concentration versus time curve from 0 to infinity, was calculated as the sum of the $AUC_{0-t} + C_{last}/k_e$. k_e is the first-order terminal constant rate calculated from a semi-log plot of a plasma concentration versus time curve. $AUC_{0-\infty}$ was taken into account when the extrapolation part was lower than 20%; due to the analytical method, no result was available for plasma data. C_{max} and t_{max} are the maximum measured plasma concentration and the time to reach it, respectively. $t_{1/2}$ is the terminal half-life, calculated as $\ln(2)/k_e$. For the calculation of the half-lives, we only considered the data above the lower limit of quantification (LLOQ = 2 $\mu g \bullet mL^{-1}$ for plasma and LLOQ = 0.1 $\mu g \bullet mL^{-1}$ for saliva).

Statistical analysis of the C_{max}, AUC_{0-t}, $AUC_{0-\infty}$, and $t_{1/2}$ was carried out with an analysis of variance, including the "HDT" and the "session" (i.e., 2001 vs. 2002) factors.[29] When a statistical difference was observed, the Newman-Keuls test was applied. Statistical analysis of the t_{max} was performed with a Kruskal-Wallis test.[29] Statistical analysis was conducted using Stata Statistical software (Stata Corporation, Texas, USA). All the statistical tests were two-sided and considered as significant for a p-value less than 0.05.

The saliva and plasma acetaminophen concentrations and pharmacokinetic parameters were compared by plotting a scattergram with a line of best fit and the calculation of the correlation coefficient (*r*).

RESULTS

Pharmacokinetic Parameters of Acetaminophen in Plasma

The mean plasma time-concentration curves of acetaminophen following the administration of 1 g during the four HDT of the 2001 and the 2002 sessions are shown in Figure 1a,b, respectively. The mean of the pharmacokinetic parameters (± *SD*) for each subject, obtained for the four HDT, are presented in Table II. The analysis of variance showed a significant increase in the mean C_{max} (30%, 48%, 83%) for HDT1, HDT18, and HDT80 compared with HDT0, respectively. The Kruskal-Wallis test indicated a significant difference between the four mean t_{max} values: t_{max} was decreased (44%, 58%, 64%) for HDT1, HDT18, and HDT80 compared with HDT0, respectively. No statistical difference was observed for the mean $AUC_{0\text{-}t}$ and $t_{1/2}$, but we noted a decreasing trend of the mean $t_{1/2}$ (9%, 18%, 31%). There was a significant difference between the 2001 and 2002 sessions for the mean $AUC_{0\text{-}t}$.

Pharmacokinetic Parameters of Acetaminophen in Saliva

As for plasma, the mean saliva time-concentration curves of acetaminophen following the administration are presented in Figure 2a,b, respectively, and the mean of the pharmacokinetic parameters (± *SD*) in Table III. For the mean C_{max}, the analysis of variance showed an increased trend for HDT1 (34%) and HDT18 (37%) and a significant increase for HDT80 (118%) compared with HDT0. The analysis of variance showed a significant increase in the mean $AUC_{0\text{-}t}$ (26%) and $AUC_{0\text{-}\infty}$ (22%) for HDT80 compared with HDT0. The Kruskal-Wallis test indicated a significant difference between the four mean t_{max}: t_{max} was decreased (47%, 51%, 70%) for HDT1, HDT18, and HDT80 compared with HDT0. No statistical difference was observed for the mean $t_{1/2}$. There was a significant difference in t_{max} between the 2001 and 2002 sessions and a borderline difference for the mean $AUC_{0\text{-}t}$ and $AUC_{0\text{-}\infty}$ (p = 0.053 and p = 0.051, respectively).

Table II Means of the Plasma Pharmacokinetic Parameters Obtained for Each Volunteer after 1 g of Acetaminophen Given to 10 (2001) and 8 (2002) Volunteers before (HDT0) and during Head-Down Tilt on Days 1, 18, and 80

Plasma	C_{max} (μg/mL)	t_{max} (h)	$AUC_{0\text{-}t}$ (μg × h/mL)	$t_{1/2}$ (h)
HDT0				
2001	6.24 ± 1.2	3.11 ± 1.14	21.18 ± 5.85	3.43 ± 2.04
2002	9.84 ± 2.66	1.89 ± 0.99	37.16 ± 6.71	2.91 ± 1.05
Global	8.04 ± 2.72	2.50[a] ± 1.2	29.17[b] ± 10.27	3.17 ± 1.58
HDT1				
2001	9.01 ± 2.86	1.47 ± 1.37	26.68 ± 6.39	2.7 ± 1.19
2002	12.21 ± 2.89	1.33 ± 1.21	39.77 ± 8.19	3.11 ± 0.69
Global	10.43[c] ± 3.23	1.41[a] ± 1.27	32.50[b] ± 9.7	2.88 ± 0.99
HDT18				
2001	11.81 ± 4.41	0.82 ± 0.38	32.22 ± 9.64	2.15 ± 0.65
2002	11.95 ± 5.46	1.25 ± 0.7	38.04 ± 7.79	2.97 ± 0.62
Global	11.89[c] ± 4.82	1.05[a] ± 0.59	35.33[b] ± 8.9	2.59 ± 0.74
HDT80				
2001	15.48 ± 5.77	1.22 ± 1.15	29.42 ± 10.25	1.85 ± 0.72
2002	13.69 ± 5.62	0.48 ± 0.17	41.12 ± 13.26	2.67 ± 0.47
Global	14.74[c] ± 5.6	0.91[a] ± 0.94	34.24[b] ± 12.66	2.19 ± 0.74

Values presented as mean ± *SD*. C_{max}, maximum observed plasma concentration; t_{max}, time to C_{max}; $AUC_{0\text{-}t}$, area under the concentration-time curve from 0 to the last measurable concentration; $t_{1/2}$, half-life.

a. Significant difference (Kruskal-Wallis test; $p < 0.05$) due to the head-down bed rest ("HDT") factor.

b. Significant difference (ANOVA; $p < 0.05$) due to the "session" factor: 2001 versus 2002.

c. Significant difference (ANOVA; $p < 0.05$) due to the "HDT" factor: HDT groups versus HDT0.

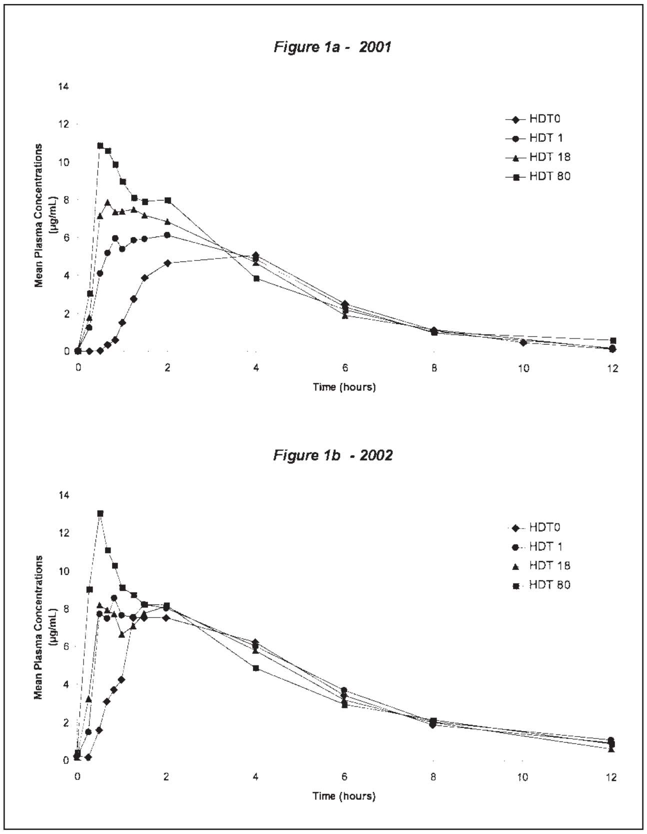

Figure 1. Mean of the plasma concentrations versus time obtained for each group of volunteers after 1 g of acetaminophen administered orally to volunteers before (HDT0) and during head-down tilt on days 1, 18, and 80: (a) 10 volunteers in 2001 and (b) 8 volunteers in 2002.

Comparison of the Saliva and Plasma Methods

Figure 3 shows a good correlation between saliva and plasma acetaminophen concentrations (354 paired samples) during 0 to 12 hours ($r = 0.8707$, $p < 0.001$). As we used two different analytical methods with two different LLOQ (HPLC: LLOQ = 0.1 $\mu g \bullet mL^{-1}$; FPIA: LLOQ = 2 $\mu g \bullet mL^{-1}$), we only considered the saliva and plasma concentrations up to 2 $\mu g \bullet mL^{-1}$.

The correlations (r) between saliva and plasma acetaminophen pharmacokinetic parameters (C_{max}, t_{max}, $AUC_{0\text{-}t}$, and $t_{1/2}$) were tested. The best correlations were observed for C_{max} ($r = 0.8313$, $n = 64$ sample pairs, $p < 0.001$) and t_{max} ($r = 0.7664$, $n = 64$ sample pairs, $p < 0.001$), two markers of gastric emptying. The correlations for the other pharmacokinetic parameters were $r = 0.5647$ ($p < 0.001$) for $AUC_{0\text{-}t}$ and $r = 0.2043$ ($p < 0.001$) for $t_{1/2}$.

DISCUSSION

The first aim of our study was to determine the influence of three durations (1, 18, and 80 days) of simulated weightlessness (–6° head-down bed rest) on gastric emptying by studying the pharmacokinetic parameters of orally administrated acetaminophen. All the plasma AUC of acetaminophen were in the same order of magnitude, revealing that the absorbed amounts were equivalent for all the bed rest periods. The unchanged plasma $t_{1/2}$ showed that the elimination was not modified. The increased plasma C_{max} and decreased plasma t_{max} were evidence of more rapid drug absorption,[21-26] which increased with the bed rest duration. Such a phenomenon might also occur with other drugs, leading to higher plasma concentrations during the absorption phase. This increase in plasma concentrations for products with a narrow therapeutic index (NTI) could lead to toxic concentrations because these particular drugs present a small difference between therapeutic and toxic concentrations. Even if at the moment, no drugs with NTI are present in the on-board medical kit,[30] these may be used in the future when many more individuals will be participating in space travel and exploration.

As there are very little data on acetaminophen plasma pharmacokinetics during spaceflights, we examined this in subjects in the simulated microgravity models of the supine position and the head-down bed rest. The horizontal supine position reproduces the same kind of cardiovascular modifications as the –6°head-down bed rest.[31] In the horizontal model, the lag time of these modifications is longer, and their intensity is less than in the head-down model.[5,6,32] Renwick et al[9] studied the influence of posture on the gastric emptying: standing, lying on the right side, and lying on the left side. Both standing and lying on the right side produced comparable delivery of acetaminophen but more rapid absorption than lying on the left side. Similarly, Anvari et al[11] observed a delayed gastric emptying with subjects lying on the left side when compared with the sitting position. Villanueva-Meyer et al[10] reported a significant delayed gastric emptying in the supine position when compared with the sitting position, but the population studied was composed of young children (ages 1 week to 4 years, 20 boys and 24 girls) who suffered from vomiting. Rumble et al[33] studied the pharmacokinetics of acetaminophen in healthy male adults on two separate occasions in the morning (during ambulation and during bed rest) and did not observe any significant decrease in t_{max}. So, the authors concluded that gastric emptying was not modified by the supine position when compared with the ambula-

Table III Means of the Saliva Pharmacokinetic Parameters Obtained for Each Volunteer after 1 g of Acetaminophen Given to 10 (2001) and 8 (2002) Volunteers before (HDT0) and during Head-Down Tilt on Days 1, 18, and 80

Saliva	C_{max} (μg/mL)	t_{max} (h)	$AUC_{0\text{-}t}$ (μg × h/mL)	$AUC_{0\text{-}\infty}$ (μg × h/mL)	$t_{1/2}$ (h)
HDT0					
2001	7.66 ± 1.47	3.28 ± 1.09	36.23 ± 7.88	39.59 ± 8.74	2.59 ± 0.41
2002	10.31 ± 3.65	2.5 ± 1.25	45.6 ± 8.04	48.38 ± 8.89	2.58 ± 0.36
Global	8.91 ± 2.96	$2.91^{a,b}$ ± 1.2	40.64 ± 9.09	43.73 ± 9.65	2.58 ± 0.37
HDT1					
2001	8.39 ± 2.31	1.9 ± 1.17	38.99 ± 8.9	41.62 ± 10.31	2.68 ± 0.4
2002	16.38 ± 4.81	1.06 ± 0.68	53.04 ± 9.75	55.87 ± 10.51	2.63 ± 0.3
Global	11.95 ± 5.39	$1.53^{a,b}$ ± 1.05	45.23 ± 11.52	47.95 ± 12.45	2.66 ± 0.35
HDT18					
2001	11.91 ± 6.42	1.7 ± 1.32	43.01 ± 10.69	45.08 ± 11.58	2.45 ± 0.4
2002	12.59 ± 5.44	1.06 ± 0.68	43.73 ± 7.3	46.78 ± 7.02	2.8 ± 0.42
Global	12.21 ± 5.84	$1.42^{a,b}$ ± 1.1	43.33 ± 9.09	45.84 ± 9.59	2.61 ± 0.43
HDT80					
2001	22.45 ± 9.95	1.1 ± 0.7	53.97 ± 7.89	55.92 ± 8.89	2.31 ± 0.44
2002	15.18 ± 7.36	0.57 ± 0.19	47.72 ± 10.85	49.96 ± 11.47	2.65 ± 0.61
Global	19.45^{c} ± 9.47	$0.88^{a,b}$ ± 0.6	51.40^{c} ± 9.45	53.47^{c} ± 10.15	2.45 ± 0.52

Values presented as mean ± *SD*. C_{max}, maximum observed plasma concentration; t_{max}, time to C_{max}; $AUC_{0\text{-}t}$, area under the concentration-time curve from 0 to the last measurable concentration; $AUC_{0\text{-}\infty}$, area under the concentration-time curve from 0 to infinity; $t_{1/2}$, half-life.
a. Significant difference (Kruskal-Wallis test; $p < 0.05$) due to the head-down bed rest ("HDT") factor.
b. Significant difference (ANOVA; $p < 0.05$) due to the "session" factor: 2001 versus 2002.
c. Significant difference (ANOVA; $p < 0.05$) due to the "HDT" factor: HDT groups versus HDT0.

tory period. These findings justified our choice of the sitting position as the reference position on HDT0. The variations of gastric emptying as a function of the posture could be explained by the pressure exerted by the stomach contents on the pylorus.

In our study, volunteers were not lying on the right or left side but on their back, and they showed a decrease in the t_{max} from the beginning to the end of the bed rest period. This decrease may be explained by a faster gastric emptying or a more rapid intestinal absorption rate. This latter hypothesis could be related to cardiovascular variations.[17,34-39] A lower total peripheral resistance has been observed after 10 minutes in the supine position compared with the sitting[36] and the standing positions,[35-37] as well as after 6 hours in the –5° head-down tilt compared with the standing position.[40] When the total peripheral resistance decreases, the blood flow increases. This is also the case for the intestinal blood flow. Acetaminophen is totally (F = 80%-90%)[41] and rapidly absorbed (t_{max} = 30-90 min)[22] through the intestinal wall by passive diffusion.[19] For this kind of drug, the rate of diffusion depends directly on the gradient of concentrations between the intestinal lumen and the intestinal blood flow.[16] Usually, the intestinal luminal concentration is higher than the intestinal blood concentration since the intestinal blood flow maintains this difference. If the intestinal blood flow increases, the intestinal blood concentration of acetaminophen will more rapidly decrease, thus accentuating the gradient of concentrations between the lumen and the blood, resulting in an increase in the absorption rate of acetaminophen. The decrease in the total peripheral resistance observed in the supine position and in head-down bed rest could be a physiological explanation of the increase in the absorption rate as well as of the decrease in the t_{max} observed in our study. However, it should be noted that the duration of the experiments describing the decrease in the total peripheral resistance in supine and head-down positions was not very long (a few minutes to a few hours). This was not the case in our study, in which the subjects were in the head-down position for much longer (80 days).

Cintron et al[42] used salivary sampling to determine changes in the concentration profiles of acetaminophen before and during spaceflight. Ground-based studies of normal subjects had established that saliva was an appropriate substitution for plasma since the saliva/plasma ratio of acetaminophen was usually around 1.[24,43,44] In the study of Cintron et al,[42] the salivary pharmacokinetics of orally ingested acetaminophen

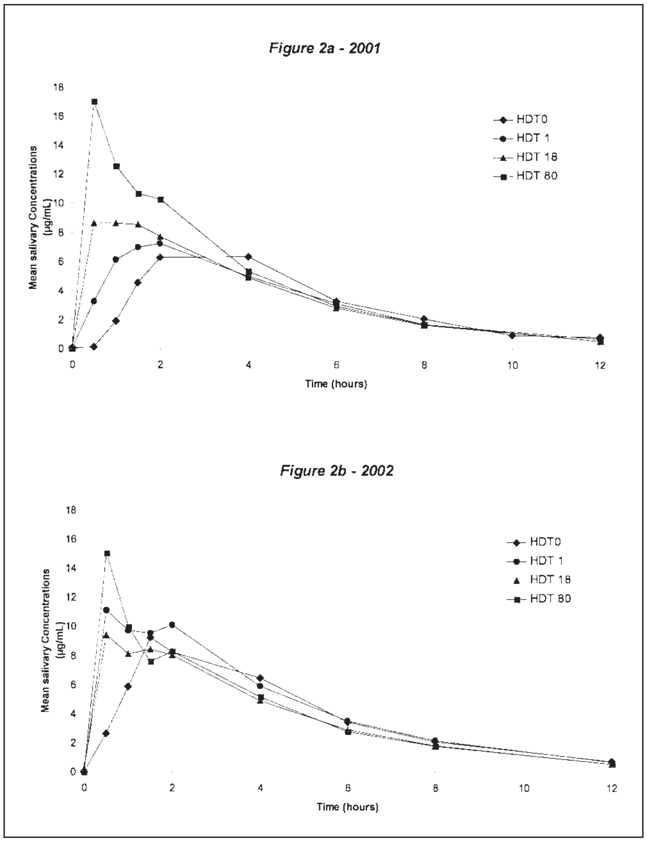

Figure 2. Mean of the saliva concentrations versus time obtained for each group of volunteers after 1 g of acetaminophen administered orally to volunteers before (HDT0) and during head-down tilt on days 1, 18, and 80: (a) 10 volunteers in 2001 and (b) 8 volunteers in 2002.

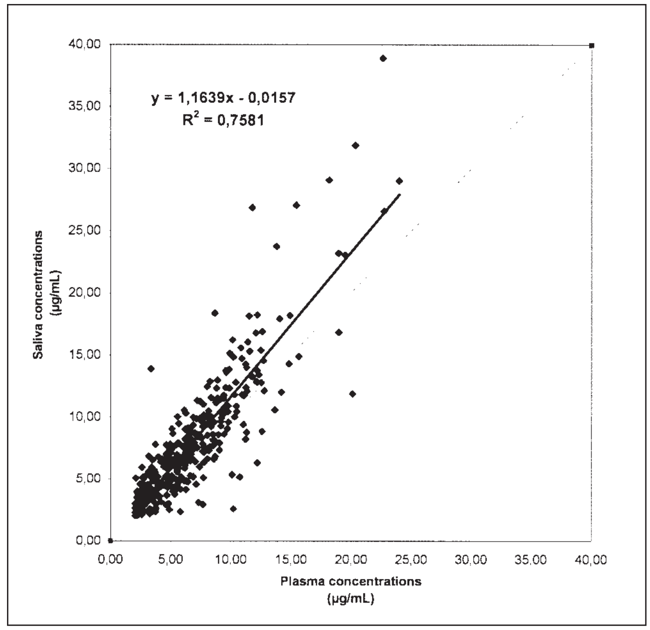

Figure 3. Linear regression of saliva and plasma acetaminophen concentrations without the head-down bed rest ("HDT") and "session" distinction.

were evaluated from salivary samples collected before and during spaceflight. The participating crewmembers were requested to collect preflight control samples. The crewmembers ingested two 325-mg tablets of acetaminophen with 100 mL of water followed by a thorough rinsing of the mouth. Saliva samples were collected at regular time intervals (15 and 30 min and 1, 2, 3, 4, 5, 6, and 8 h) after dosing. The concentration of acetaminophen in each sample was determined later using a rapid HPLC analysis method. Cintron et al emphasized that the results were preliminary because of the small number of subjects tested in flight and the variations in investigation characteristics between the flights. Data obtained from 5 crewmembers on three different missions showed significant changes in the concentration profiles of acetaminophen during flight. When compared with the preflight control, the absorption phase of the profile exhibited a more pronounced change than the elimination phase. The C_{max} and the t_{max} seemed to vary according to the time in flight, although no subjects were tested more than once. The intersubject variability in the C_{max} and the t_{max} was low preflight but higher during spaceflight. This may have been because of problems of adaptation, including vomiting due to space motion sickness. Data from 2 crewmembers, on day 2 of the mission, indicated that the time to reach peak concentration was faster and that higher peak drug concentrations were obtained than in the ground-based studies. Data from 2 crewmembers on day 4 of the mission showed a significant decrease in the absorption of acetaminophen (C_{max} decreased and t_{max} increased) compared with the preflight profile. The C_{max}, as well as the t_{max} and its disposition, may be influenced by space motion sickness,[3] which occurred in 1 crewmember. In addition, other artifacts, such as inadequate rinsing of the mouth, may also have been contributing factors.[3] Physiological adaptation to weightlessness may reach an early equilibrium on day 4, and this may explain the decreased absorption of acetaminophen. We could not perform a valid comparison with our results for two main reasons. First, the high variability of the data and the lack of intraflight comparison were a limiting factor. In addition, the salivary sampling method used by Cintron et al involved stimulating salivary flow before sampling. This was not done in the current study as nonstimulated saliva data had been shown to give good correlations with plasma,[24,43,44] and it was also felt that it might produce more variability of sampling.

The significant differences observed between the 2001 and 2002 sessions for the mean saliva t_{max} and the mean plasma AUC remain unexplained since the acetaminophen batch, the diet, fluid intake, physical exercise, and experimental protocol were identical during the two sessions. Given that no explanation could be found, the correct statistical procedure in this study was to test the influence of the session on the pharmacokinetic parameters and to report the statistical results.

The second aim of this study was to compare saliva and plasma results. A good correlation ($r = 0.8707$) between saliva and plasma concentrations was observed. These results are in agreement with those obtained by Smith et al.[43] In that study, following the oral administration of 1.5 g of acetaminophen, blood and saliva samples were collected simultaneously at 30 and 60 minutes. Smith et al found a significant correlation between saliva and plasma concentrations. Similarly, after an oral administration of 1.5 g of acetaminophen to healthy volunteers, Kamali et al[44] found a significant correlation between plasma and saliva levels during the latter phase of the experiment (i.e., 65-360 min). We also observed a good correlation between saliva and plasma pharmacokinetic parameters ($r = 0.8313$ for C_{max}, $r = 0.7664$ for t_{max}, $r = 0.5647$ for AUC_{0-t}). These results are corroborated by those obtained by Sanaka et al.[24] In their study, the subjects consumed 20 mg/kg of acetaminophen by the oral route, and there was a good correlation for C_{max}, t_{max}, and $AUC_{1.0\ h}$ between plasma and saliva.

The good correlations between the saliva and plasma concentrations and pharmacokinetic parameters may warrant the substitution of saliva sampling for blood sampling in the acetaminophen method. During spaceflight, the saliva method, being noninvasive and easy to perform, even if the analytical method is longer (HPLC vs. FPIA), could be reliably used for routine pharmacokinetic studies.[20]

In conclusion, we have observed that during a –6° head-down bed rest of 80 days, there was a significant increase in the C_{max} and a significant decrease in the t_{max}, from the beginning to the end of the bed rest period, without modification of the amount of drug absorbed. A possible explanation was that there was a decrease in the total peripheral resistance observed in head-down tilt or accelerated gastric emptying. The good correlation between plasma and saliva results indicated that it should be possible to use saliva sampling during spaceflight: in-flight pharmacokinetic results will or will not validate our simulated microgravity model used to study absorption.

The simulated weightlessness induced an increase in plasma concentrations during the absorption phase (increase in C_{max}). For drugs with an NTI, this increase could lead to toxic effects. While NTI drugs are not currently used during space missions, the situation may be different in the future with larger numbers of people engaging in long spaceflights and exploration. Recommendations for the use of such drugs will probably have to include a reduction in dose to compensate for the pharmacokinetic changes seen during prolonged weightlessness.

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